

Mohammed Omar Khan

SYSTEMS PLATFORM ENGINEER · MLOps

Abu Dhabi, UAE

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MLOps engineer focused on turning ML experimentation into reliable, observable production systems.

Summary

Systems and platform engineer with 3+ years of experience across infrastructure, data engineering, automation and machine learning. Currently operate production platforms inside a strictly network-isolated UAE government environment, spanning Red Hat OpenShift, Nutanix, Linux/Solaris, Python/Bash automation and centralized observability. Outside work, build and operate a self-hosted Kubernetes platform for ML workloads, including offline LLM serving through vLLM, RAG, agentic workflows, persistent storage, CI/CD and Prometheus/Grafana observability. Seeking to apply this infrastructure and ML background to industrialize the machine-learning lifecycle from experimentation through reliable, observable deployment.

Experience

Midis Group

Abu Dhabi, UAE

SYSTEMS ENGINEER – DEPLOYED ONSITE TO A UAE GOVERNMENT AGENCY

Mar 2024 – Present

- Operate as the onsite technical point of contact within a strictly air-gapped government environment, translating operational requirements into deployed and supportable solutions.
- Administer Nutanix and Red Hat OpenShift platforms, mixed Linux/Solaris servers and a local Elastic Cloud Enterprise environment, with support for GPU-equipped infrastructure.
- Designed, deployed and operate centralized Elasticsearch + Kibana observability used by stakeholders for operational visibility and troubleshooting.
- Author Python and Bash automation for operational data and reporting workflows; use Ansible for infrastructure automation and OpenShift Pipelines for platform workflows.
- Used n8n to automate operational workflows, improving workplace efficiency by at least 20%.
- Coordinate with vendor teams and troubleshoot across operating-system, application, network and platform layers in a constrained production environment.

MDS TSS (Midis Group)

UAE

BIG DATA INTERN

May 2023 – Mar 2024

- Completed Hadoop training and contributed to customer big-data engagements, including cluster deployment exercises and pipeline tuning.
- Developed Python automation for customer data-mining and analysis workflows and produced operational reports covering ticket throughput and SLA performance.

Heriot-Watt University Dubai

Dubai, UAE

TEACHING ASSISTANT

Sep 2022 – May 2023

- Mentored 50+ Computer Science students in Python, machine-learning algorithms, data visualization, Jupyter and ML tooling.
- Assisted faculty across undergraduate Computer Science modules, including AI/ML-focused courses.

Selected MLOps / Engineering Projects

Self-Hosted ML Platform

LOCAL GENAI STACK – LLM SERVING, RAG AND AGENTIC ORCHESTRATION

- Deployed Qwen, DeepSeek and LLaMA-family open-weight models fully offline on a self-managed Kubernetes cluster, benchmarking vLLM, Hugging Face TGI and Ollama as inference runtimes.
- Exposed an OpenAI-compatible inference API through vLLM and configured tool calling with automatic tool choice and a Hermes-style parser for agentic applications.
- Built LangGraph orchestration workflows with stateful multi-step execution, tool selection, conditional routing and handoff patterns.
- Ran GGUF, GPTQ and AWQ quantized models and measured throughput/latency trade-offs across runtimes and quantization levels to select suitable deployment configurations for constrained GPU resources.
- Built a LangChain RAG pipeline over a Markdown knowledge base using local embeddings and Chroma, tuning chunking, retrieval top-K and prompt construction against context-window constraints.

Kubernetes / Proxmox / Observability

SELF-HOSTED KUBERNETES PLATFORM

- Built and operate a 3-node kubeadm Kubernetes cluster across Proxmox and bare-metal nodes for stateful services and AI workloads.
- Designed storage using dynamic persistent-volume provisioning through CSIs, CloudNativePG for stateful PostgreSQL workloads and backups.
- Implemented cluster networking and access control, with Tailscale for controlled remote connectivity to applications.
- Deployed kube-prometheus-stack, Grafana, Loki and Alloy for metrics and log observability across cluster and application workloads.
- Run Forgejo as self-hosted Git infrastructure, supporting reproducible deployment and Git-centric operational workflows.

Python / GitHub Actions / PyPI

VSCAN – OPEN-SOURCE PYTHON SECURITY SCANNER

- Led development of a Python-based scanner integrating external APIs to analyze files and websites for malicious content.
- Built a GitHub Actions CI/CD pipeline that runs tests on pull requests and automatically builds and publishes versioned PyPI releases from tagged commits.

PyTorch / TensorFlow / Scikit-learn / NLP

ADVERTISEMENT DETECTION – FINAL-YEAR CAPSTONE

- Implemented and trained classifiers on a 1M+ row URL dataset, engineering NLP-based features from URL structure to detect advertising links.
- Compared neural networks, XGBoost, SVMs, LightGBM and Random Forests, achieving 80% classification accuracy.

Technical Skills

MLOps / ML	MLflow, ML lifecycle, model serving, experiment tracking, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM
LLMs / GenAI	vLLM, TGI, Ollama, Hugging Face, quantization, LangGraph, LangChain, RAG, Chroma, tool calling, MCP
Containers / Platform	Kubernetes, Docker, Red Hat OpenShift, OpenShift Pipelines, Proxmox, Helm, Nutanix
CI/CD / Automation	Git, GitHub Actions, Forgejo, Ansible, Terraform, Python, Bash, n8n
Observability / Data	Prometheus, Grafana, Loki, Alloy, Elasticsearch, Kibana, ECE, Pandas, NumPy, SQL, ETL, Hadoop
APIs / Software	Python, FastAPI, Flask, REST APIs, microservices, PyPI packaging
Operating Systems	Linux (RHEL, Rocky, Ubuntu, Debian), Solaris

Education

Heriot-Watt University Dubai

Dubai, UAE

B.Sc. (Hons) COMPUTER SCIENCE WITH ARTIFICIAL INTELLIGENCE – UPPER SECOND CLASS HONOURS (2:1)

Sep 2020 – Jun 2023

- Deputy Principal's Award for academic excellence.
- AI-focused coursework included neural networks, supervised and unsupervised machine learning, applied text analysis/NLP and computer vision foundations.

Certifications

- 2024** Data Science A-Z – SuperDataScience
- 2023** Azure Data Fundamentals – Microsoft
- 2023** Career Essentials in Data Analysis – Microsoft
- 2021** Certified Junior Associate – Java – Oracle

Languages

Languages English (fluent), Hindi, Urdu